

AP CALCULUS AB - UNIT 4

Guided Notes 4.1: Interpretations of the Derivative



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we interpret a derivative value as an instantaneous rate of change in context?

- Interpret a derivative value as an instantaneous rate of change in context.
- State derivative units by dividing output units by input units.
- Use the sign and magnitude of a derivative to describe local behavior.
- Distinguish the meaning of a quantity, its first derivative, and its second derivative.

Key Knowledge, Vocabulary, and Notation

Contextual derivative

If $Q = Q(t)$, then $Q'(a)$ is the instantaneous rate at which Q changes with respect to t at $t = a$.

Units

Units of $Q'(t)$ are output units per input unit; units of $Q''(t)$ are output units per input-unit squared.

Sign

$Q'(a) > 0$ means Q is increasing locally; $Q'(a) < 0$ means it is decreasing locally.

Magnitude

A larger $|Q'(a)|$ means faster instantaneous change, not necessarily a larger quantity.

Complete interpretation

Name the quantity, input value, direction, numerical rate, and units.

Second derivative

$Q''(a)$ describes how the first rate Q' is changing at the instant.

Concept Development and Strategy

1. Read the notation before calculating: identify the dependent quantity, the independent variable, and their units.
2. Do not say that $Q'(a)$ is the amount of Q . It is the instantaneous rate of change of Q at input a .
3. A negative derivative describes direction of change; it does not imply the original quantity is negative.
4. For a table, $Q'(a)$ may be estimated using a symmetric difference when data on both sides are available.
5. Interpret $Q''(a)$ as the rate of change of Q' ; include squared input units in the denominator.

Shared Representation / Data

Quantity	Input	Derivative	Interpretation pattern
$P(t)$ people	t years	$P'(t)$ people/year	population change at time t
$C(q)$ dollars	q items	$C'(q)$ dollars/item	marginal cost near output q
$T(h)$ °C	h hours	$T'(h)$ °C/hour	temperature change at hour h
$V(r)$ cm ³	r cm	$V'(r)$ cm ²	sensitivity of volume to radius

Worked Example: Population interpretation

A city population $P(t)$ is measured in thousands of people, where t is years after 2025. Interpret $P'(4) = 2.7$.

Answer and Reasoning

In 2029, the city population is increasing at an instantaneous rate of 2.7 thousand people per year.

Worked Example: Marginal cost

The cost $C(q)$ is measured in dollars for producing q units. Interpret $C'(600) = 14.20$.

Answer and Reasoning

Near a production level of 600 units, total cost is increasing by approximately \$14.20 for each additional unit produced.

Worked Example: Second derivative

Temperature $T(t)$ is in degrees Celsius and time is in minutes. Interpret $T''(8) = -0.35$.

Answer and Reasoning

At minute 8, the rate at which temperature changes is decreasing by 0.35°C per minute each minute.

Worked Example: Estimate from data

A medication concentration satisfies $C(2.9) = 18.41$ and $C(3.1) = 17.73$ mg/L. Estimate and interpret $C'(3)$.

Answer and Reasoning

$C'(3) \approx \frac{17.73 - 18.41}{0.2} = -3.4$ mg/L/min. At 3 minutes, concentration is decreasing at about 3.4 mg/L each minute.

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. The mass $M(t)$ is in grams and t is in seconds. State the units of $M'(t)$ and $M''(t)$.

2. Interpret $V'(12) = -5.8$ when V is liters and time is minutes.

3. Revenue $R(q)$ is in dollars. Interpret $R'(120) = 38$.

4. A lake depth $D(t)$ is in meters and $D'(5) = -0.012$. Write a complete contextual statement.

5. Two quantities satisfy $A'(3) = -12$ and $B'(3) = 7$. Which is changing faster at 3? Explain.

6. A pollutant amount $A(t)$ satisfies $A(6) = 90$, $A'(6) = -4$, and $A''(6) = 1.2$. Give a complete local description.

7. Let $Q(t) = f(g(t))$. Write a contextual interpretation of $Q'(a) = f'(g(a))g'(a)$ in words.

Multiple-Choice Check

1. If $H(t)$ is height in meters and t is seconds, the units of $H''(t)$ are

- ## Common Errors and AP Precision

- ## Exit Ticket

A model gives $P'(2) = 0$ but measurements show P increasing on both sides of 2. Explain how this is possible.